Seeing Through Sunlight: Single-Photon Avalanche Diode Cameras for Space Applications

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***Abstract*—Vision-based pose estimation is a fundamental capability for space operations, but challenging illumination geometries and high-dynamic-range scenes can cause conventional cameras to lose visual information through saturation and underexposure. Single-Photon Avalanche Diode (SPAD) cameras offer a promising alternative imaging modality, detecting individual photons at high frame rates while providing higher dynamic range than conventional sensors. To support the exploration of SPAD imagery, we introduce PocketSPAD (available: https://zenodo.org/records/19075296), a dataset of synchronized SPAD and RGB image sequences of a 5×5×5 cm PocketQube satellite captured under diverse illumination conditions, including direct sun-facing scenes. We show that SPAD imagery retains substantially more usable features than RGB under saturating sunlight and robustly yields more usable frames under diverse lighting conditions. Aggregation of low-bit SPAD images into higher bit-depths improves pairwise feature matching at the cost of temporal resolution. Motivated by this trade-off, we propose a lightweight autoregressive convolutional neural network–gated recurrent unit (CNN–GRU) pipeline for continuous 6-DoF pose estimation. We demonstrate that it matches or exceeds the pose-estimation accuracy of more traditional Kalman filter-based methods, remains robust under simulated occlusion and blackout, exhibits stable long-sequence inference, and achieves high-rate inference on the tested GPUs. Interestingly, pose-estimation accuracy does not consistently benefit from higher bit-depths, instead often favoring the higher temporal resolution of lower bit-depths under simulated bit-dependent sampling. These results highlight the potential of SPAD cameras for space applications.**

***Index Terms*—Space Robotics and Automation, Deep Learning for Visual Perception, Data Sets for Robotic Vision**

I. Introduction

SPACE applications such as in-orbit proximity operations, planetary landing, rover path planning, and earth observation increasingly exploit vision-based sensors. The respective environments routinely feature low diffusivity and span several orders of magnitude in illumination requiring conventional CMOS and CCD sensors to work at their operational limits, often eroding the spatial features needed for tasks such as object detection or 6 Degree-of-Freedom (6-DoF) pose estimation [1], [2]. For safety, or to increase the chances of success, missions employing CMOS or CCD cameras routinely impose operational limitations such as attitude constraints to avoid sun-facing scenarios [3], [4] or operational windows to work in specific lighting conditions [5].

In the particular case of uncooperative autonomous 6-DoF spacecraft pose estimation, there is a requirement for: high fidelity feature extraction, sufficient processing power to regress or track the target pose, and online operation in the case of rendezvous applications. Conventional solutions employed by Oumer and Panin [6] and Rondao et al. [7] used stereo and monocular cameras respectively and used feature extractors, a 6-DoF pose initialization and Kalman filters.

Single-photon avalanche diode (SPAD) cameras provide an alternative sensing mechanism, being able to detect individual photons at frame rates exceeding 100k FPS and operate in low-light or sun-facing scenes, providing high temporal resolution while representing high dynamic ranges across challenging illumination conditions [8], as illustrated in the left panel of Fig. 1.

Motivated by these properties, this paper investigates the imaging characteristics of SPAD cameras and their potential for space-domain visual tasks. We use feature matching and continuous 6-DoF spacecraft pose estimation as use cases, focusing on challenging illumination conditions and exploring the high frame rates and different bit depths of SPAD images.

The main contributions of this work are threefold:

1) We introduce PocketSPAD, a laboratory dataset containing both SPAD and conventional RGB images of a small PocketQube in diverse lighting conditions.
2) We compare RGB and SPAD imagery, highlighting the advantages of SPAD sensing in challenging illumination.
3) Motivated by the high temporal sampling and multi-bit depth exploration, we propose a lightweight autoregressive convolutional neural network–gated recurrent unit (CNN–GRU) estimator for efficient continuous 6-DoF spacecraft pose estimation, presented in Fig. 1.

II. Background and Related Work

A SPAD differs from other photodiodes in that the diode is biased slightly above its breakdown voltage (i.e., Geiger mode or overvoltage). With some probability, a single photon can trigger an electron avalanche that produces sufficient current

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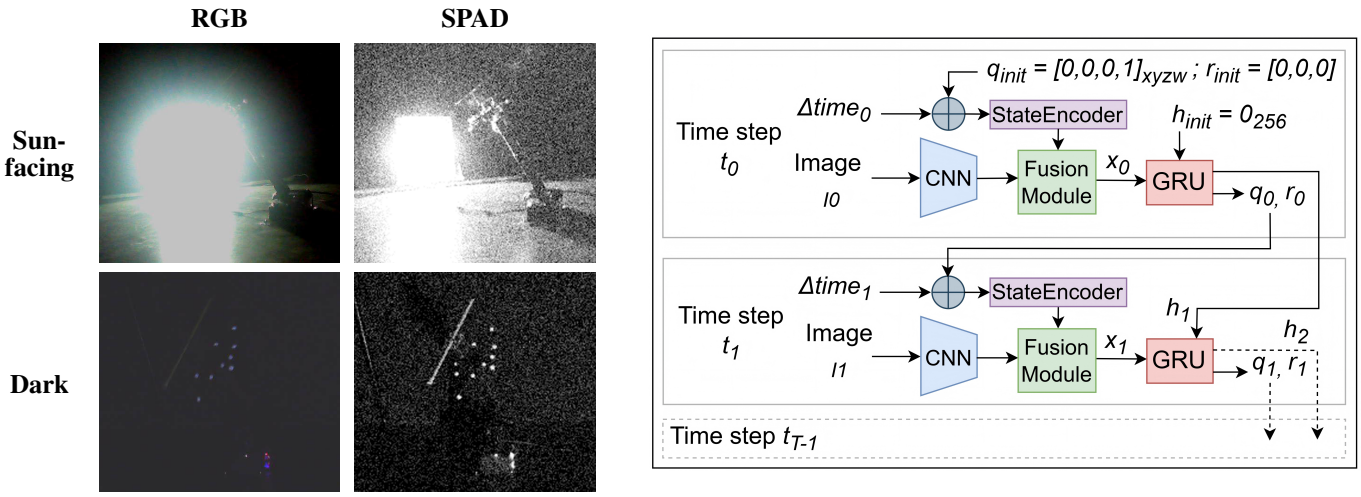


Fig. 1: Left: Representative RGB and SPAD observations from PocketSPAD under dark and sun-facing conditions. Right: Proposed autoregressive CNN–GRU workflow. At each time step t_k, where $1 \leq k \leq T-1$, the image I_k, the previously predicted pose (q_{k-1}, r_{k-1}), and the inter-frame interval Δtime_k between I_{k-1} and I_k are used to predict the pose (q_k, r_k).

to indicate a detection event. A quenching circuit subsequently stops the avalanche and recharges the device. During the resulting dead time, the pixel is insensitive to new photon arrivals [8]. Both **imaging** and **event** modality SPADs are available: the pixel activations can be stored either in a binary tensor (i.e., similar to a conventional image) or as a time sequence of $[x, y]_{pixel}$ photon detection events (i.e., similar to an event camera). In the image-based (i.e., **binary tensor**) modality considered here, each pixel records a binary value indicating whether at least one photon detection occurred during an exposure window [9]. Higher-bit-depth intensity frames are obtained through temporal aggregation. In the SPAD512 camera used in this work, an n-bit frame is formed by summing $2^n - 1$ consecutive binary frames (e.g., a 4-bit frame aggregates 15 binary frames and takes values from 0 to 15) [10]. This aggregation improves the photon-count signal-to-noise ratio (SNR) and intensity resolution, but reduces the output frame rate, resulting in a trade-off between photon-count fidelity and temporal resolution [11].

Frequently, space missions have employed cameras in monocular/stereo/panoramic configurations with visible-light/multispectral/infrared sensors for lunar rovers [12], [13], Earth observation [14], and spacecraft rendezvous and docking [15]. Event cameras offer high dynamic range [16] and have been investigated for spacecraft vision, including sim-to-real adaptation for satellite pose estimation [1], [17], [18]. However, their reliance on temporal contrast can result in sparse data during periods of low relative motion [19]. In contrast, SPAD cameras detect photon arrivals, extracting features even in stationary, low contrast or high illuminance conditions [9], [11]. Existing space-domain datasets rely almost entirely on RGB imagery such as SPEED, SPEED+, SwissCube, Hayabusa2, Soyuz/Dragon, and PocketQube [20]–[25], or event-based data like SEENIC, SPADES, and EVOS [17], [26], [27]. In contrast, the use of SPADs for space-based vision remains largely unexplored. To our knowledge, only one SPAD dataset exists for space applications, targeting lunar ego-motion: SPICE-HL3 [28].

Deep neural networks have been widely explored for uncooperative spacecraft pose estimation [29], with keypoint-based pipelines predicting spacecraft landmarks and recovering pose via geometric solvers [22], [30]. Their effectiveness has been demonstrated through the Satellite Pose Estimation Challenges conducted using SPEED and SPEED+ [20], [31], with subsequent work targeting computationally efficient deployment on resource-constrained hardware [32]–[34]. However, single-frame methods do not exploit temporal information available from image sequences. The classic approach to incorporating temporal context is model-based filtering; Kalman filters have been combined with spacecraft pose estimates for kinematic state tracking [6], [7], [35], with ESKFs particularly well-suited to states involving 3D rotations. Learned temporal models offer an alternative: recurrent architectures such as GRUs [36] capture sequential dependencies directly from data and have been applied to visual localization [37], though their use for spacecraft pose tracking remains limited. To explore the trade-off between SPAD temporal resolution and image-feature fidelity, we employ a lightweight autoregressive CNN–GRU estimator for continuous 6-DoF pose estimation from high-rate, low-bit SPAD image sequences, and further evaluate it against CNN–MLP–ESKF baselines to explore the potential of SPAD cameras for space applications.

III. Custom Datasets

To help address the scarcity of available SPAD datasets, we introduce PocketSPAD, which contains image sequences of a 5x5x5cm PocketQube satellite attached to the end-effector of a robotic arm imaged simultaneously with a conventional RGB camera (the ORBBEC Astra Embedded S) and a SPAD camera (Pi Imaging Technologies' SPAD 512). The RGB and SPAD cameras provide spatial resolutions of 1920×1080 and 512×512 pixels, respectively. The lighting conditions include ambient lighting, darkness, and a collimated light source (i.e., sunlight simulator) placed to approximate different orbital

TABLE I: Overview of the PocketSPAD sequences. Long and short denote relative recording durations, whereas N, NW, and SE denote sun-simulator positions relative to the camera viewpoint. A dash denotes unavailable data.

Illum.	Variant	SPAD acq. fmt.	SPAD	RGB
			frames / FPS / dur. (s)	frames / FPS / dur. (s)
Ambient	Full	4-bit	35,800 / 3635.17 / 9.848	35 / 3.23 / 10.520
Ambient	Partial long	1-bit	866,250 / 1589.21 / 545.080	1,707 / 8.37 / 203.759
Ambient	Partial short	1-bit	39,525 / 2160.96 / 18.290	422 / 10.61 / 39.676
Dark	Exp. 1	4-bit	4,150 / 400 / 10.373	41 / 3.63 / 11.020
Dark	Exp. 2	4-bit	6,800 / 666.67 / 10.199	–
Dark	Exp. 3	4-bit	2,718 / 285.71 / 9.510	–
Sun_N	Long	–	–	607 / 8.90 / 68.069
Sun_N	Short	4-bit	29,250 / 3125.00 / 9.520	136 / 14.02 / 9.628
Sun_NW	–	1-bit	1,154,745 / 2122.73 / 543.990	3,928 / 7.19 / 545.954
Sun_SE	Long	1-bit	1,002,480 / 1839.26 / 545.044	3,121 / 5.70 / 547.521
Sun_SE	Short Exp. 1	–	–	22 / 2.12 / 9.905
Sun_SE	Short Exp. 2	4-bit	26,100 / 2631.58 / 10.062	–
Sun_SE	Short Exp. 3	4-bit	27,800 / 2631.58 / 10.311	–
Total	–	–	**3,195,618 / – / 1722.227**	**10,019 / – / 1446.052**

lighting scenarios. The robotic arm executed pre-generated kinematic trajectories. Ground-truth 6-DoF target poses were recorded at 30 Hz using an OptiTrack system and associated with the RGB and SPAD image frames through nearest-timestamp matching. Each pose annotation describes the target pose in the coordinate frame of the corresponding camera, with rotation represented by a scalar-last quaternion (q_x, q_y, q_z, q_w) and translation by a Cartesian vector (r_x, r_y, r_z).

PocketSPAD sequences were acquired natively as either 1-bit binary or 4-bit photon-count frames. 1-bit firmware and data-throughput constrained long sequences combined with short exposures, thus some sequences were recorded in 4-bit mode. Native 1-bit frames were captured in 255-frame batches using sequence-dependent exposure times, with processing intervals between batches. For a common 4-bit representation, consecutive groups of 15 native 1-bit frames were accumulated within each batch. Table I summarizes the 13 PocketSPAD sequences. RGB and SPAD sequences within the same ambient-partial recording may differ in duration, due to either SPAD buffer overflow or, in one case, an early RGB termination. Frames affected by acquisition errors were removed but we still report the full sequence duration. PocketSPAD is publicly available at https://zenodo.org/records/19075296.

IV. Method

A. RGB and SPAD Feature-Matching Protocol

To compare the spatial information retained in RGB and SPAD imagery under challenging lighting conditions, we evaluate RGB and 4-bit SPAD sequences from PocketSPAD using the frozen pretrained XFeat detector and descriptor [38] without modality-specific fine-tuning. We evaluate the `dark_exp1` and `sun_NW` sequences, representing underexposure and saturation, respectively. For each modality, an axis-aligned region enclosing the target and its attached markers is manually selected once in the initialization frame forming a fixed template. Each subsequent frame is matched independently against this template. RGB is evaluated at its native acquisition rate. For SPAD, we evaluate with two protocols: (1) SPAD-rate, using every available frame, and (2) RGB-rate, subsampling the frames to approximately match the RGB rate.

TABLE II: Effective frame rates after temporal aggregation of the four native 1-bit PocketSPAD sequences.

Sequence	1-bit	2-bit	4-bit	8-bit
`sun_NW`	2122.73	707.58	141.52	8.32
`sun_SE_long`	1839.26	613.09	122.62	7.21
`ambient_partial_long`	1589.21	529.74	105.95	6.23
`ambient_partial_short`	2160.96	720.32	144.06	8.47

We report the mean counts of mutual and pose-consistent matches per frame, along with the count and percentage of usable frames. First, mutual nearest-neighbor matches quantify bidirectionally consistent descriptor associations. Second, the object-to-camera pose annotations are used to construct an annotation-derived essential matrix between the initialization and query frames. After lens-distortion correction and conversion to normalized camera coordinates, each mutual match is evaluated against this geometry using its symmetric point-to-epipolar-line angular error. A match is considered pose-consistent when this error is at most 0.1°. Finally, a frame is considered usable when it contains at least five pose-consistent matches.

B. SPAD Bit-Depth Evaluation Protocol

To assess spatial information retention across different SPAD bit depths and their associated temporal-aggregation settings, we evaluate several feature-tracking algorithms on 1-, 2-, 4-, and 8-bit representations from the native 1-bit PocketSPAD and SPICE-HL3 sequences; we report the best performing method here, ORB [39]. Each n-bit input is formed by accumulating $2^n - 1$ consecutive binary frames. Consecutive frames in each representation are matched using ORB as a non-learned reference, with Lowe-ratio filtering and RANSAC verification; the reported inlier ratio is the fraction of retained matches classified as RANSAC inliers. Because this procedure simultaneously increases bit depth and reduces effective frame rate, the observed ORB performance reflects their combined influence rather than an isolated bit-depth effect. For PocketSPAD, the corresponding effective frame rates are f_1, $f_1/3$, $f_1/15$, and $f_1/255$, respectively, where f_1 is the sequence-specific effective native 1-bit frame rate. Table II reports the resulting frame rates. The per-pair matching metric characterizes the usability of individual frame pairs, but does not establish whether higher frame rate observations at lower bit depths benefit tasks that integrate information over time. This motivates pose estimation in the following section for probing the temporal–fidelity trade-off.

C. Pose-Estimation Implementation

We compare a learned recurrent estimation approach (CNN–GRU) with a pose estimation coupled with filtering approach (CNN–MLP–ESKF) to examine alternative ways of exploiting temporal information in SPAD image sequences. Both methods employ ResNet-18 [40] as the backbone. We consider the ESKF pipelines as representative of more conventional pose-estimation approaches.

1) Pose-Estimation Pipelines: The **CNN–GRU pipeline** is shown in the right-hand panel of Fig. 1. The pose input is initialized with an identity pose and the GRU hidden state to zero. At each subsequent time step t_k, the pipeline receives the current image I_k, the preceding pose prediction (q_{k-1}, r_{k-1}), and the inter-frame time interval Δtime_k. The CNN extracts image features, while the state encoder embeds the preceding pose and time interval. Their fused representation is passed to the GRU, which updates its hidden state and predicts the current pose (q_k, r_k). During training, sequences are divided into windows of length T using a fixed stride, and predictions are fed back autoregressively within each window. During evaluation, the hidden state and pose predictions are propagated over the complete sequence without ground-truth pose updates or intermediate resets. The GRU has an input dimension of 256, a hidden-state dimension of 256, and one recurrent layer, containing 395 thousand trainable parameters.

The **CNN–MLP–ESKF (AR) pipeline** follows the same workflow as the CNN–GRU pipeline, but replaces the GRU with a comparably sized MLP with layer dimensions 256–512–256–256–256 and 394 thousand trainable parameters, followed by an ESKF. The ESKF is used only during evaluation and does not participate in network training. During training, the unfiltered MLP prediction is fed back autoregressively. During evaluation, the MLP prediction is treated as the ESKF measurement input, and the ESKF pose estimation is fed back as the pose input at the next time step. Here, AR denotes the autoregressive training strategy.

The **CNN–MLP–ESKF (SF) pipeline** follows the CNN–MLP–ESKF (AR) baseline but removes the autoregressive pose feedback and state encoder. The MLP therefore predicts its pose measurement solely from the current image, after which the prediction is processed by the ESKF during evaluation. The fusion module is retained to map the visual features into a higher-dimensional representation. Here, SF denotes the single-frame neural prediction strategy.

2) Data Split: All available sequences other than `ambient_full`, `dark_exp1`, and `sun_SE_short_exp2` were assigned to the training set. The validation/test sets were partitioned into contiguous temporal segments to study temporal performance. Specifically, `dark_exp1` and `sun_SE_short_exp2` were each divided chronologically into two equal halves, with the first half used for validation and the second half used for testing. For `ambient_full`, the first 50% of the sequence was used for training, the following 25% for validation, and the final 25% for testing. As related motion profiles could recur across sequences, this protocol evaluates representative illumination conditions rather than generalization to unseen trajectories or lighting.

3) Training Configuration: Neural pose estimators were trained with the Adam optimizer (seed 42, learning rate 10^{-4}, batch size 32) for up to 100 epochs, with an early stopping patience of 3. Sequences were divided into windows of length 32 (stride 4) and shuffled before training. Translation is guided by a relative ℓ_2 loss (ε ensures numerical stability):

$$\mathcal{L}_{\text{trans}} = \frac{\|\hat{\mathbf{r}} - \mathbf{r}\|_2}{\|\mathbf{r}\|_2 + \varepsilon}. \tag{1}$$

TABLE III: Effective input frame rates under bit-dependent temporal subsampling.

Sequence	1-bit	2-bit	4-bit
`ambient_full`	3635.17	1211.72	242.34
`dark_exp1`	400.00	133.33	26.67
`sun_SE_short_exp2`	2631.58	877.19	175.44

Rotation is guided with an algebraic quaternion loss that accounts for the antipodal equivalence $\mathbf{q} \equiv -\mathbf{q}$:

$$\mathcal{L}_{\text{rot}} = \frac{\min(\|\hat{\mathbf{q}} - \mathbf{q}\|_2,\ \|\hat{\mathbf{q}} + \mathbf{q}\|_2)}{\|\mathbf{q}\|_2 + \varepsilon}. \tag{2}$$

Despite the broader discontinuity associated with direct four-dimensional rotation representations [41], we observed stable training convergence with no evidence of representation-related instability. The final training loss uniformly sums the translation and rotation losses.

4) Evaluation Metrics: For evaluation, translation error followed the same relative ℓ_2 formulation as $\mathcal{L}_{\text{trans}}$, while rotation error was measured as the angular distance in degrees between the predicted and ground-truth quaternions:

$$e_{\text{rot}} = \frac{180}{\pi}\, 2 \arccos\big(\text{clip}\big(\big|\hat{\mathbf{q}}^\top \mathbf{q}\big|, 0,\ 1\big)\big). \tag{3}$$

D. Pose-Estimation Protocol

1) Temporal–Fidelity Evaluation: To evaluate the temporal–fidelity trade-off, we derive 1-, 2-, and 4-bit inputs from the native 4-bit PocketSPAD sequences. Given $I_4 \in \{0, \ldots, 15\}$, we derive $I_1 = \mathbf{1}[I_4 > 7.5]$ and $I_2 = \lfloor I_4/4 \rfloor$, while retaining I_4. The 1-, 2-, and 4-bit inputs are normalized by their bit-depth maxima (1, 3, and 15, respectively) to $[0, 1]$, and a separate model is trained for each bit depth. During evaluation, the 1-, 2-, and 4-bit inputs are temporally subsampled using strides of 1, 3, and 15, respectively, approximating the frame-rate reduction of the $2^n - 1$ binary exposures required to form an n-bit frame. The resulting frame rates are reported in Table III.

2) Robustness Evaluation: During training, occlusion (a randomly placed rectangle covering 50% of each dimension) and blackout (an all-zero or all-one frame) are randomly applied to PocketSPAD images with probabilities of 10% and 5%, respectively. During evaluation, each corruption is applied separately to a fraction $\rho \in \{10\%, 20\%, \ldots, 90\%, 95\%, 98\%, 100\%\}$ of frames, using corruption seeds $\{42, 43, 44\}$.

3) Inference Speed Evaluation: We used CUDA Graphs and automatic mixed precision to reduce inference overhead, measuring sequentially at batch size one over 10 runs on an A100-SXM4-80GB (cluster) and an RTX 2060 (laptop). Per-frame latency (mean $\pm$ std) excludes image loading, preprocessing, and CPU–GPU transfer to isolate estimator computation from implementation-dependent overhead.

4) Drift Evaluation: To evaluate long-sequence drift, we run the GRU and ESKF (AR) pipelines over each complete 4-bit test sequence without reinitialization. Per-frame errors are plotted as 100-step causal moving averages.

5) Learning Validation: To isolate the contributions of motion priors versus visual observations: **Persistence** holds the first ground-truth pose throughout; **Causal constant-velocity** extrapolates linear and angular velocity from two distinct ground-truth poses; **Image-free** replaces all SPAD images with zero tensors at evaluation, without fine-tuning.

E. ESKF Implementation

1) Kalman Filter: The ESKF maintains a nominal pose $\hat{\mathbf{x}}_k = (\hat{\mathbf{q}}_k, \hat{\mathbf{r}}_k)$ and a 6D error state $\delta\mathbf{x}_k = [\delta\boldsymbol{\theta}_k^\top \ \delta\mathbf{r}_k^\top]^\top$, where $\delta\boldsymbol{\theta}_k$ is a local tangent-space rotation error and $\delta\mathbf{r}_k$ is an additive Euclidean translation error. To avoid unreliable adjacent-frame differences at the high SPAD frame rates, the constant velocities are estimated over $W = 64$ frames:

$$\mathbf{v}_k = \frac{\hat{\mathbf{r}}_k - \hat{\mathbf{r}}_{k-W}}{t_k - t_{k-W}}, \qquad \boldsymbol{\omega}_k = \frac{2\,\mathrm{vec}\big(\hat{\mathbf{q}}_{k-W}^{-1} \otimes \hat{\mathbf{q}}_k\big)}{t_k - t_{k-W}}, \tag{4}$$

where t_k and t_{k-W} are the current and window-start timestamps, respectively. The error covariance is propagated as

$$\mathbf{P}_{k+1} = \mathbf{F}_k \mathbf{P}_k \mathbf{F}_k^\top + \mathbf{Q}\Delta t_k, \tag{5}$$

where $\mathbf{F}_k$ is the linearized transition matrix and $\mathbf{Q}$ is the process-noise covariance.

We compute rotation and translation residuals from the CNN–MLP output as

$$\mathbf{y}_k = \begin{bmatrix} 2\,\mathrm{vec}\big(\hat{\mathbf{q}}_k^{-1} \otimes \mathbf{q}_{\mathrm{meas},k}\big) \\ \mathbf{r}_{\mathrm{meas},k} - \hat{\mathbf{r}}_k \end{bmatrix}. \tag{6}$$

The standard Kalman update produces the correction $\delta\mathbf{x}_k$, which is injected as

$$\hat{\mathbf{q}}_k \leftarrow \hat{\mathbf{q}}_k \otimes \mathrm{Exp}(\delta\boldsymbol{\theta}_k), \qquad \hat{\mathbf{r}}_k \leftarrow \hat{\mathbf{r}}_k + \delta\mathbf{r}_k. \tag{7}$$

All other steps follow the standard ESKF formulation.

2) ESKF Hyperparameter Calibration: The measurement covariance is calibrated from the unfiltered CNN–MLP predictions on the equally weighted 4-bit validation sequences. Let $\bar{e}_{t,\mathrm{rel}}$ and $\bar{e}_\theta$ denote the averages of the sequence-level relative translation and rotation errors in radians, respectively, and let $\|\mathbf{r}\|_{\mathrm{rep}}$ denote the equally weighted average of the sequence-level mean ground-truth translation magnitudes. To reduce sensitivity to trajectory-specific directional biases, we use an isotropic measurement-covariance model and set

$$\sigma_t = \|\mathbf{r}\|_{\mathrm{rep}} \bar{e}_{t,\mathrm{rel}},\ \sigma_\theta = \bar{e}_\theta,\ \mathbf{R} = \mathrm{diag}(\sigma_\theta^2 \mathbf{I}_3, \sigma_t^2 \mathbf{I}_3). \tag{8}$$

Because the pose is initialized as the identity pose, we set $\mathbf{P}_0 = 10\mathbf{R}$ to represent its high initial uncertainty. The process covariance is parameterized as $\mathbf{Q} = \lambda\mathbf{R}$, with λ selected by validation loss from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. We restrict $\lambda \leq 1$ because the CNN–MLP measurements are expected to be less reliable than the motion prior. Occlusion and blackout are calibrated separately at $\rho = 0.5$, and used across all corruption levels.

Based on the representative translation magnitude $\|\mathbf{r}\|_{\mathrm{rep}} = 2010\,\mathrm{mm}$ computed from the validation sequences, table IV reports the covariance parameters selected on the validation set. The clean configuration is calibrated for the CNN–MLP–ESKF (AR) baseline for use in the subsequent drift evaluation (Section V-C5). Relative to the clean AR configuration, the corruption-specific configurations use larger measurement covariances and smaller λ values. This is consistent with placing greater reliance on the ESKF motion prediction as the neural measurements become less reliable.

TABLE IV: ESKF covariance parameters calibrated using the 4-bit SPAD validation sequences.

Pipeline	Condition	R_θ (rad^2)	R_r (mm^2)	Selected λ
AR	Clean	1.0114×10^{-3}	26.890	10^{-1}
AR	Occlusion	3.632×10^{-3}	130.368	10^{-3}
AR	Blackout	5.739×10^{-3}	184.128	10^{-2}
SF	Occlusion	3.930×10^{-2}	1269.145	10^{-4}
SF	Blackout	7.321×10^{-2}	2165.774	10^{-4}

Notes: The corruption-specific parameters were calibrated once using corruptions generated with seed 42, avoiding seed-specific tuning.

TABLE V: Quantitative comparison of RGB and 4-bit SPAD feature matching using a frozen XFeat model.

Condition	Stream	Frames	$\frac{Matches}{Frame}$ ↑	$\frac{PC}{Frame}$ ↑	Usable frames: Count ↑	Usable frames: (%) ↑
dark_exp1	RGB	40	23.50	**5.43**	9	22.50
	SPAD (RM)	40	39.55	2.90	9	22.50
	SPAD (native)	4,142	**40.84**	3.11	**944**	**22.79**
sun_NW	RGB	3,927	32.20	1.28	343	8.73
	SPAD (RM)	3,849	**41.32**	1.92	594	15.43
	SPAD (native)	76,966	**41.32**	**1.96**	**12,214**	**15.87**

Notes: RM denotes the rate-matched SPAD stream, with sampling steps of 102 and 20 for `dark_exp1` and `sun_NW`, respectively. Matches and PC denote mutual and pose-consistent matches, respectively. Frames exclude the initialization frame and queries with degenerate epipolar geometry.

V. EXPERIMENTS

A. RGB versus SPAD Feature Matching Results

RGB and SPAD feature-matching results are shared in Table V. Under `sun_NW`, SPAD yields more mutual and pose-consistent matches per frame and a higher percentage of usable frames, consistent with the loss of spatial information caused by RGB saturation. Under `dark_exp1`, SPAD (RM) yields more matches, but with a lower average pose-consistent matches per frame. However SPAD (RM) exhibited the same usable-frame percentage. This hints that the RGB modality contains richer features but it is worth noting that was trained with XFeat's conventional RGB images. At native acquisition rate, SPAD provides substantially more geometrically usable observations under both conditions. This denser stream may benefit continuous pose estimation during fast relative motion, though PocketSPAD was not designed to evaluate this and we leave it for future work.

B. SPAD Bit-Depth Evaluation Results

Fig. 2 shows representative ORB feature matching for sun_SE_long, while Table VI summarizes the results across all sequences in PocketSPAD and SPICE-HL3. The inlier ratio consistently increases with bit depth, with the largest gains occurring between 1 and 4 bits. Increasing from 4 to 8 bits provides only marginal improvement, indicating that 4 bits retain most of the matching performance of 8 bits. Notably,

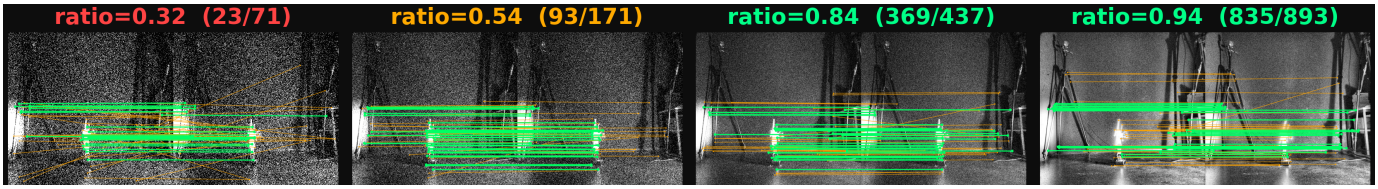


Fig. 2: ORB feature matching on `sun_SE_long` across 1-, 2-, 4-, and 8-bit depths (left to right).

TABLE VI: ORB inlier ratio (Lowe-ratio + RANSAC). † indicates no scene structure.

Dataset	Condition	1-bit	2-bit	4-bit	8-bit
PocketSPAD	sun_NW	0.327	0.663	0.848	0.913
	sun_SE_long	0.330	0.709	0.859	0.941
	amb_p_long	0.444	0.780	0.917	0.962
	amb_p_short	0.439	0.772	0.913	0.966
SPICE-HL3	dawn-on	0.061	0.223	0.442	0.536
	dawn-off	0.093	0.367	0.598	0.673
	noon-on	0.066	0.230	0.454	0.579
	noon-off	0.079	0.318	0.541	0.625
	night-on	0.068	0.145	0.325	0.413
	night-off†	-	-	-	-

ORB failed to extract meaningful features in the pitch-black SPICE-HL3 night-off sequence, instead erroneously reporting dark current noise as features.

C. Pose-Estimation Results

1) Temporal–Fidelity: We use the proposed CNN–GRU pipeline to evaluate the temporal–fidelity trade-off in terms of pose-estimation accuracy. Table VII reports the quantitative results, and Fig. 3 contains 4-bit qualitative examples.

No single bit-depth and sampling-rate combination performs best across all sequences: 1-bit is best under ambient illumination (`ambient_full`), 2-bit under sunlight (`sun_SE_short_exp2`), and both 1- and 2-bit outperform 4-bit under darkness (`dark_exp1`). Within the simulated sampling protocol, this suggests a promising advantage of the SPAD modality: exploiting different bit depths for different contexts. Viewed alongside the ORB results, the pose-estimation results show that better pairwise matching at higher bit depths does not necessarily translate into lower pose errors under the respective protocols.

2) Robustness Evaluation: Figure 4 Left compares the three pipelines under increasing occlusion and blackout. The CNN–GRU exhibits the strongest overall robustness across most evaluated corruption levels, with its errors remaining nearly constant under occlusion and under blackout up to 98% corruption rates. This stability reflects the temporal propagation between informative observations. Together with the high SPAD acquisition rate and small inter-frame motion, this allows the CNN–GRU to bridge corrupted intervals. The CNN–MLP–ESKF (AR) pipeline also consistently outperforms its single-frame (SF) counterpart under occlusion and under moderate-to-high blackout, indicating that the filtered pose feedback provide useful temporal information when visual evidence is partially degraded. Under near-complete blackout, however, both autoregressive pipelines degrade sharply while the single-frame baseline degrades more gradually, likely reflecting recursive error accumulation: with fixed identity initialization and little informative input, erroneous predictions are repeatedly fed back into the pipeline. The mean standard deviation bands across runs are negligible, indicating low sensitivity to the sampled corruption pattern.

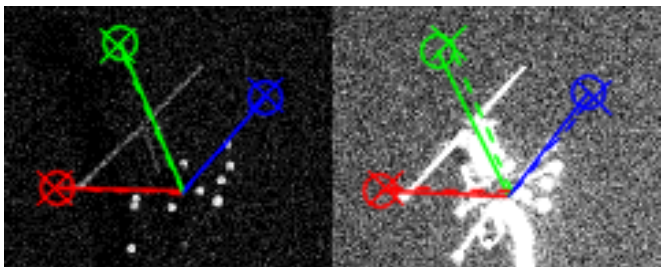


Fig. 3: CNN–GRU qualitative performance on the 4-bit PocketSPAD `dark_exp1` and `sun_SE_short_exp2`. GT: solid axes with circles; Pred.: dashed axes with crosses.

TABLE VII: CNN–GRU pose-estimation performance on PocketSPAD under simulated bit-dependent frame-rate reduction. Bold denotes the lowest error across bit-depth settings for each illumination condition and metric.

Bit Depth	Trans. Error (relative; mm, ↓)			Rot. Error (°, ↓)		
	amb_full	dark1	SE_s2	amb_full	dark1	SE_s2
1	**0.0036 (7.2)**	**0.0020 (4.0)**	0.0073 (14.7)	**1.47**	1.21	8.58
2	0.0043 (8.6)	0.0023 (4.6)	**0.0035 (7.0)**	1.82	**1.15**	**1.90**
4	0.0043 (8.6)	0.0058 (11.7)	0.0035 (7.0)	1.82	2.98	1.99

Notes: The millimetre equivalents of the relative translation errors are computed using $\|\mathbf{r}\|_{\text{rep}} = 2010\,\text{mm}$.

TABLE VIII: Inference latency on PocketSPAD reported as mean ± standard deviation ($\frac{ms}{frame}$)

Pipeline	Cluster		Laptop	
	Latency ↓	FPS ↑	Latency ↓	FPS ↑
GRU	$\mathbf{1.427 \pm 0.012}$	**700.77**	$\mathbf{4.001 \pm 0.178}$	**249.94**
MLP–ESKF (AR)	1.885 ± 0.034	530.50	4.930 ± 0.236	202.84
MLP–ESKF (SF)	1.763 ± 0.039	567.22	4.768 ± 0.229	209.73

3) Inference Speed Evaluation: Table VIII reports the per-frame inference time and FPS of each pipeline on the cluster and laptop platforms. The CNN–GRU pipeline achieved the fastest inference speeds, reaching 1.427 ms/frame (700 FPS) on the cluster and 4.001 ms/frame (250 FPS) on the laptop.

4) Drift Evaluation: Both pipelines maintain bounded errors without systematic growth across test sequences. Figure 4

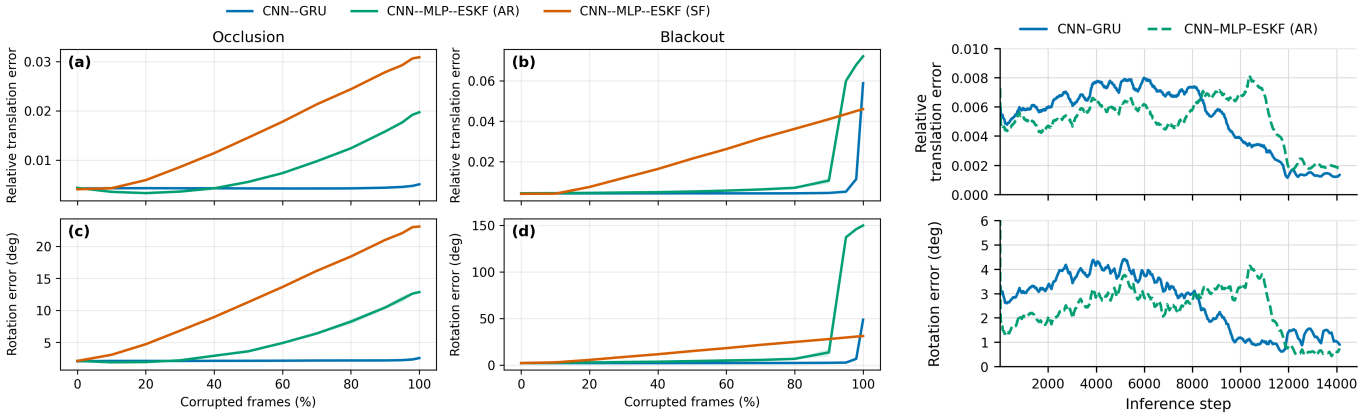


Fig. 4: **Left:** Robustness to occlusion and blackout (solid: unweighted mean across sequences; shading: ± standard deviation). **Right:** Long-sequence autoregressive errors on `sun_SE_short_exp2`. Both evaluations use 4-bit SPAD data.

TABLE IX: Motion-prior baselines, image-free ablation, and best CNN–GRU performance across bit depths.

Method	Trans. Error (relative; mm, ↓)			Rot. Error (°, ↓)		
	amb_full	dark1	SE_s2	amb_full	dark1	SE_s2
Persistence	0.031 (62.3)	0.054 (108.5)	0.048 (96.5)	16.85	29.22	26.47
Const. velocity	0.005 (10.1)	0.019 (38.2)	0.012 (24.1)	2.29	4.70	3.55
Image free	0.063 (126.6)	0.061 (122.6)	0.054 (108.5)	49.79	50.13	48.01
CNN–GRU	**0.0036 (7.2)**	**0.0020 (4.0)**	**0.0035 (7.0)**	**1.47**	**1.15**	**1.90**

(right) shows the error evolution on `sun_SE_short_exp2`, the longest test sequence, spanning approximately 14,000 inference steps under a challenging sun-facing condition.

5) Learning Validation: Table IX compares the motion-prior and image-free ablations. Persistence and image-free estimation produce substantially higher errors, whereas constant velocity has a much reduced error. This highlights a limitation of the PocketSPAD dataset: the robotic-arm positional trajectories are limited and provide a strong motion prior. A higher DoF setup would be beneficial for future datasets.

6) Hyperparameter Ablation: Table X contains a summary of ablation studies performed during CNN–GRU development. The final architecture employed a hidden dimension of 256, a training window length of 32, and a stride of 4.

TABLE X: Development-stage ablation study of the GRU hidden dimension and temporal training parameters on the 4-bit SPAD validation sequences.

Study	Setting	Trans. error (relative; mm, ↓)	Rot. error (°, ↓)
Hidden dimension	$H = 64$	0.012 (24.1)	10.49
	$H = 128$	**0.007 (14.1)**	7.45
	$H = 256^{\dagger}$	**0.007 (14.1)**	**5.96**
Window length (87.5% overlap)	$T = 16,\ S = 2$	**0.004 (8.0)**	4.35
	$T = 32^{\dagger},\ S = 4$	**0.004 (8.0)**	**3.15**
	$T = 64,\ S = 8$	0.007 (14.1)	5.96
Stride	$T = 32,\ S = 4^{\dagger}$	**0.004 (8.0)**	**3.15**
	$T = 32,\ S = 8$	0.007 (14.1)	6.70
	$T = 32,\ S = 16$	0.010 (20.1)	6.30

Notes: Errors are averaged with equal weight assigned to each sequence. H, T, and S denote the GRU hidden dimension, training-window length, and stride, respectively. The † symbol denotes the selected configuration.

VI. Conclusions and Future Work

This paper introduces PocketSPAD, one of the first SPAD modality datasets, to support the greater community in the investigation of SPAD imaging for feature matching and 6-DoF pose estimation in space applications. Compared with RGB sensors, SPAD imagery was seen to offer improved feature matching under sunlight and saturation conditions, as well as providing more frequent usable observations in all lighting conditions.

It was observed that temporal aggregation (higher bit-depths) improved ORB feature matching for any two sequential frames. However, under simulated bit-dependent sampling, the proposed pose-estimation pipeline favoured higher framerates at the expense of a lower-bit depth, depending on the illumination conditions. This highlights the potential adaptability of the SPAD sensor for missions involving different lighting conditions. In practice, deployment should also account for relative motion and onboard computing constraints when selecting the temporal–fidelity trade-off.

Finally, to accommodate the varying bit-depths and high frame rates of SPADs, a CNN–GRU pose-estimation pipeline was proposed in this paper. The CNN-GRU pipeline exhibited similar or better pose-estimation tracking to a more conventional feature-fed Kalman filter pipeline.

Future work could expand PocketSPAD or provide a new dataset with more native 1-bit recordings and more complex, diverse, and faster relative-motion trajectories. Additionally, online experiments measuring end-to-end latency would help assess practical real-time performance. Other future directions include evaluating higher-bit, lower-rate SPAD imagery against RGB in conventional filtering and applications such as SLAM, trajectory- and illumination-disjoint evaluations, sim-to-real transfer, and direct SPAD–event camera comparisons. We hope that PocketSPAD and these findings will support further research into SPAD imaging and contribute to more robust visual perception systems for future space missions.

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REFERENCES

[1] N. Munasinghe, C. Le Gentil, J. Naylor, M. Asavkin, D. G. Dansereau, and T. Vidal-Calleja, "Towards event-based satellite docking: A photometrically accurate low-earth orbit hardware simulation," in *HERMES2 Workshop, IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.

[2] L. Pasqualetto Cassinis, R. Fonod, E. Gill, I. Ahrns, and J. Gil Fernandez, "Comparative assessment of image processing algorithms for the pose estimation of uncooperative spacecraft," in *Proc. 10th Int. Workshop Satell. Constellations Formation Flying (IWSCFF)*, 2019, paper IWSCFF 19-43.

[3] R. Gallon, F. Schiemenz, A. Menicucci, and E. Gill, "Addressing camera sensors faults in vision-based navigation: Simulation and dataset development," *Acta Astronaut.*, vol. 239, pp. 849–872, 2026.

[4] E. B. Bierhaus *et al.*, "The OSIRIS-REx spacecraft and the touch-and-go sample acquisition mechanism (TAGSAM)," *Space Sci. Rev.*, vol. 214, no. 7, 2018, Art. no. 107.

[5] C. Zhang, P. Wu, X. Li, J. Li, and R. Zhao, "Front-lit cooperative rendezvous window rapid planning under dynamic illumination constraints," *Aerosp. Sci. Technol.*, vol. 178, pt. B, 2026, Art. no. 112500.

[6] N. W. Oumer and G. Panin, "Tracking and pose estimation of non-cooperative satellite for on-orbit servicing," in *Proc. Int. Symp. Artif. Intell., Robot. Autom. Space (i-SAIRAS)*, 2012.

[7] D. Rondao, N. Aouf, M. A. Richardson, and V. Dubanchet, "Robust on-manifold optimization for uncooperative space relative navigation with a single camera," *J. Guid. Control Dyn.*, vol. 44, no. 6, pp. 1157–1182, 2021.

[8] C. Bruschini, H. Homulle, I. M. Antolovic, S. Burri, and E. Charbon, "Single-photon avalanche diode imagers in biophotonics: review and outlook," *Light Sci. Appl.*, vol. 8, no. 1, 2019, Art. no. 87.

[9] A. C. Ulku *et al.*, "A 512×512 SPAD image sensor with integrated gating for widefield FLIM," *IEEE J. Sel. Topics Quantum Electron.*, vol. 25, no. 1, pp. 1–12, 2019, Art. no. 6801212.

[10] Pi Imaging Technology, "cSPAD system manual," https://piimaging.com/doc-cspad, 2026, ver. 2.07. Accessed: 2026-09-16.

[11] S. Ma, S. Gupta, A. C. Ulku, C. Bruschini, E. Charbon, and M. Gupta, "Quanta burst photography," *ACM Trans. Graph.*, vol. 39, no. 4, pp. 79:1–79:16, 2020.

[12] J. Wang *et al.*, "Computer vision in the teleoperation of the Yutu-2 rover," *ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. V-3-2020, pp. 595–602, 2020.

[13] J.-F. Yang *et al.*, "Panoramic camera on the Yutu lunar rover of the Chang'e-3 mission," *Res. Astron. Astrophys.*, vol. 15, no. 11, pp. 1867–1880, 2015.

[14] M. Drusch *et al.*, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sens. Environ.*, vol. 120, pp. 25–36, 2012.

[15] M. Pyrak and J. Anderson, "Performance of Northrop Grumman's mission extension vehicle (MEV) RPO imagers at GEO," in *Proc. SPIE*, vol. 12115, 2022, Art. no. 121150A.

[16] Z. Liu, S. Liang, B. Guan, Y. Shang, Q. Yu, and J. Zhao, "A geometric framework for absolute pose and velocity estimation with event cameras," *IEEE Trans. Image Process.*, vol. 35, pp. 6714–6729, 2026.

[17] M. Jawaid, E. Elms, Y. Latif, and T.-J. Chin, "Towards bridging the space domain gap for satellite pose estimation using event sensing," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023, pp. 11 866–11 873.

[18] M. Jawaid, R. Talak, Y. Latif, L. Carlone, and T.-J. Chin, "Test-time certifiable self-supervision to bridge the Sim2Real gap in event-based satellite pose estimation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2024, pp. 4534–4541.

[19] Z. Wu, M. Gehrig, Q. Lyu, X. Liu, and I. Gilitschenski, "LEOD: Label-efficient object detection for event cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16 933–16 942.

[20] M. Kisantal, S. Sharma, T. H. Park, D. Izzo, M. Märtens, and S. D'Amico, "Satellite pose estimation challenge: Dataset, competition design, and results," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 5, pp. 4083–4098, 2020.

[21] T. H. Park, M. Märtens, G. Lecuyer, D. Izzo, and S. D'Amico, "SPEED+: Next-generation dataset for spacecraft pose estimation across domain gap," in *Proc. IEEE Aerosp. Conf. (AERO)*, 2022, pp. 1–15.

[22] Y. Hu, S. Speierer, W. Jakob, P. Fua, and M. Salzmann, "Wide-depth-range 6D object pose estimation in space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 15 865–15 874.

[23] A. Price and K. Yoshida, "A monocular pose estimation case study: The Hayabusa2 Minerva-II2 deployment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2021, pp. 1992–2001.

[24] P. F. Proença and Y. Gao, "Deep learning for spacecraft pose estimation from photorealistic rendering," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2020, pp. 6007–6013.

[25] N. Sajjad, A. Price, M. Mirshams, and M. Salzmann, "Tiny satellites, big challenges: A feasibility study of machine vision pose estimation for PocketQubes during conjunctions," *Acta Astronaut.*, vol. 228, pp. 1023–1041, 2025.

[26] A. Rathinam, H. Qadadri, and D. Aouada, "SPADES: A realistic spacecraft pose estimation dataset using event sensing," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 11 760–11 766.

[27] A. Crain and S. Ulrich, "EVent-based observation of spacecraft (EVOS): A neuromorphic dataset for spacecraft proximity operations," Federated Research Data Repository, 2025, data set. [Online]. Available: https://doi.org/10.20383/103.01538

[28] D. Rodríguez-Martínez, D. van der Meer, J. Song, A. Bera, C. J. Pérez-del-Pulgar, and M. A. Olivares-Mendez, "SPICE-HL3: Single-photon, inertial, and stereo camera dataset for exploration of high-latitude lunar landscapes," *Sci. Data*, vol. 13, no. 1, 2026, Art. no. 374.

[29] L. Pauly, W. Rharbaoui, C. Shneider, A. Rathinam, V. Gaudillière, and D. Aouada, "A survey on deep learning-based monocular spacecraft pose estimation: Current state, limitations and prospects," *Acta Astronaut.*, vol. 212, pp. 339–360, 2023.

[30] B. Chen, J. Cao, A. Parra, and T.-J. Chin, "Satellite pose estimation with deep landmark regression and nonlinear pose refinement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2019, pp. 2816–2824.

[31] T. H. Park *et al.*, "Satellite pose estimation competition 2021: Results and analyses," *Acta Astronaut.*, vol. 204, pp. 640–665, 2023.

[32] Z. Wang, J. Wang, J. Yu, Z. Li, and Q. Yu, "High-accuracy real-time satellite pose estimation for in-orbit applications," *Chin. J. Aeronaut.*, vol. 38, no. 6, 2025, Art. no. 103458.

[33] P. Ancey, A. Price, S. Javed, and M. Salzmann, "FastPose-ViT: A vision transformer for real-time spacecraft pose estimation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2026, pp. 7873–7882.

[34] J. Posso, G. Bois, and Y. Savaria, "Real-time spacecraft pose estimation using mixed-precision quantized neural network on COTS reconfigurable MPSoC," in *Proc. 22nd IEEE Interregional NEWCAS Conf.*, 2024, pp. 358–362.

[35] F. Renis, A. L. Price, M. Battipede, S. Hellmich, and M. Salzmann, "Leveraging light-curve inversion for kinematic state estimation of uncooperative targets," in *Proc. 9th Eur. Conf. Space Debris*, vol. 9, no. 1, 2025.

[36] K. Cho *et al.*, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 1724–1734.

[37] S. Kim, I. Kim, L. F. Vecchietti, and D. Har, "Pose estimation utilizing a gated recurrent unit network for visual localization," *Appl. Sci.*, vol. 10, no. 24, 2020, Art. no. 8876.

[38] G. Potje, F. Cadar, A. Araujo, R. Martins, and E. R. Nascimento, "XFeat: Accelerated features for lightweight image matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 2682–2691.

[39] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *Proc. Int. Conf. Comput. Vis. (ICCV)*, 2011, pp. 2564–2571.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[41] Y. Zhou, C. Barnes, J. Lu, J. Yang, and H. Li, "On the continuity of rotation representations in neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 5738–5746.